

Proximity Sensors

Autonomous Navigation of Mobile Robots

Anna Semakova

a.semakova@gmail.com

20.02.2025

Sensors of Wheeled Robots

Perception of the environment

Active (based on time of flight or phase shift)

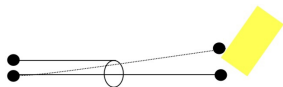
- Ultrasound
- Laser range finder
- Infrared

Passive (intensity-based)

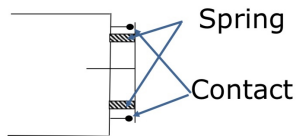
- Cameras
- Tactiles

Tactile Sensors

Measure contact with objects



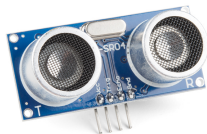
Touch sensor



Bumper sensor

Ultrasound Sensors

- Emit an ultrasound signal
- Wait until they receive the echo
- Time of flight sensor



HC-SR04

Time of Flight Sensors



$$d = \frac{v \cdot t}{2}$$

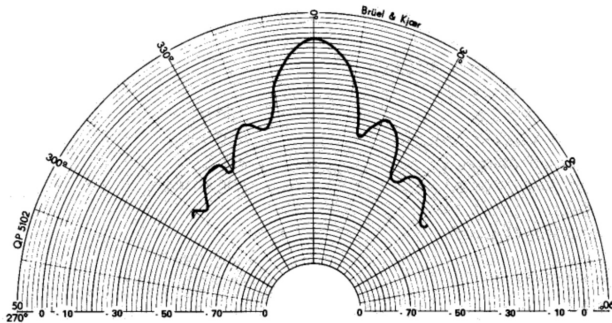
v – speed of the signal

t – time elapsed between broadcast of signal and reception of the echo

Properties of Ultrasounds

Signal Profile

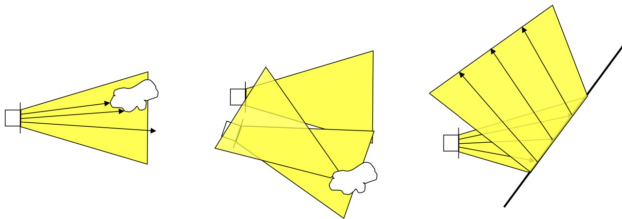
Polaroid 6500



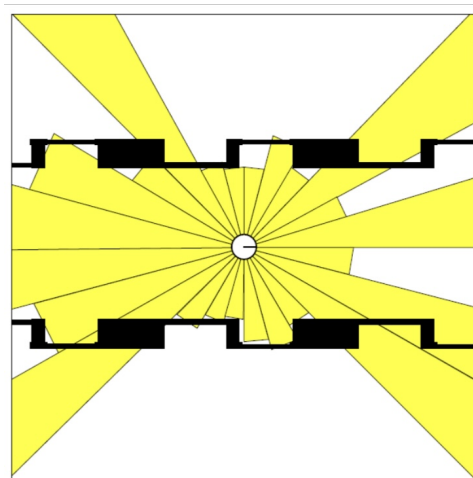
Properties of Ultrasounds

Sources of Error

- Opening angle
- Crosstalk
- Specular reflection



Typical Ultrasound Scan

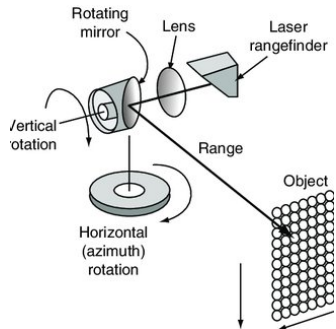


Parallel Operation

- Given a 15° opening angle, 24 sensors are needed to cover the whole 360° area around the robot
- Let the maximum range we are interested in be 10 m
- The time of flight then is $2 \cdot 10$ m divided by the speed of sound (330 m/sec) which leads to 0.06 sec
- A complete scan thus requires $24 \cdot 0.06 = 1.45$ sec
- To allow frequent updates (necessary for high speed) the sensors have to be fired in parallel
- This increases the risk of crosstalk

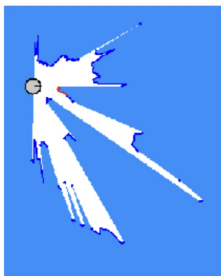
Laser Range Scanner

- High precision
- Wide field of view
- Some laser scanners are security approved for emergency stops (collision detection)



Computing the End Points

- Laser data comes as an array of range readings, e.g. $[1; 1.2; 1.5; 0.1; 81.9; \dots]$
- Assume an field of view of 180°
- First beams starts at $-1/2$ of the FoV
- Maximum range ~ 80 m (SICK LMS)



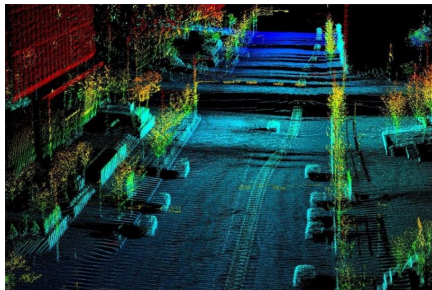
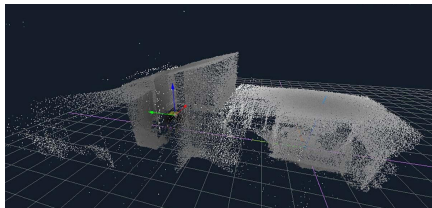
Computing the End Points

Typical problems to be solved:

- Where are the end points relative to the sensor location?
- Where are the end points in an external coordinate system?

Typical Scans

LiDAR



Another Range Sensor

Kinect

